
THE HIDDEN ENGINE

505,000 km³
per year

Today

-
- 01 **Scale**
How much water Earth holds — and where
 - 02 **Stress**
Three centuries of measurement and change
 - 03 **Pathways**
Where every drop goes after it falls
 - 04 **Response**
Three levers that work
-



THE HIDDEN ENGINE

More energy than every power plant on Earth — combined.

WHY WATER MATTERS

The hydrological cycle drives weather, agriculture, and freshwater supply for eight billion people. It moves heat from the tropics to the poles and shapes every ecosystem on the planet.

505K

KM³ WATER CYCLED PER YEAR

= 16 Mt evaporated every second

1,300

EJ LATENT HEAT TRANSPORTED

vs ~600 EJ all human energy

ACT 1 — SCALE

Four reservoirs

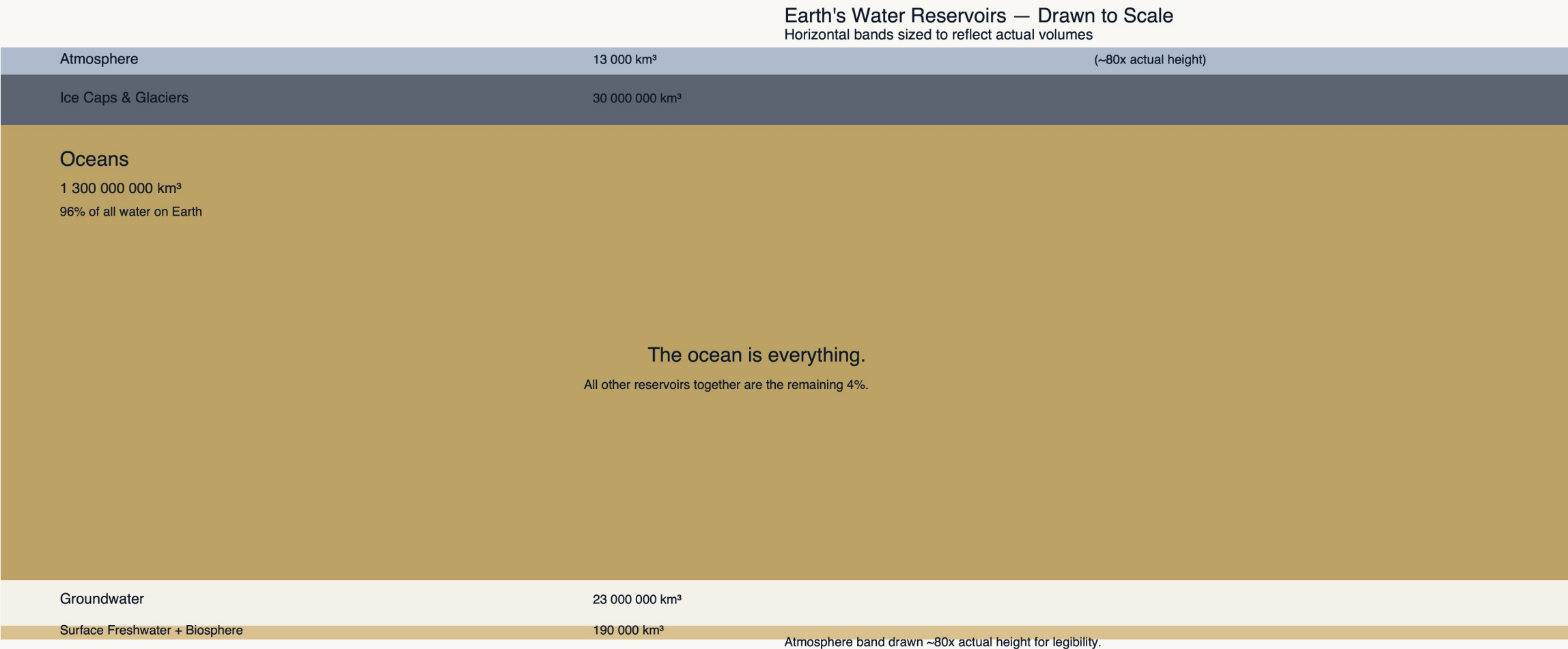


So what: Atmosphere holds a rounding error vs. the ocean — yet drives all weather.



ACT 1 — SCALE

Drawn to scale



ACT 1 — SCALE

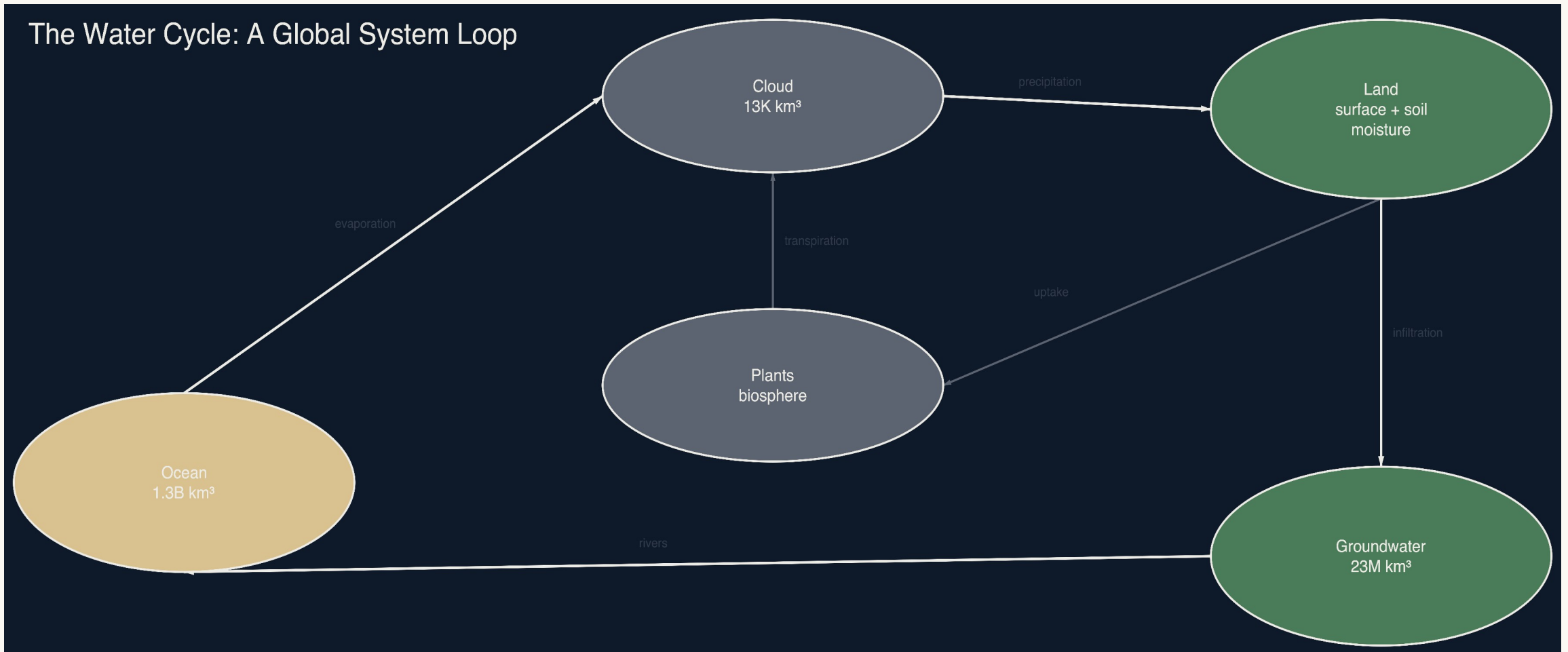
01

Earth's Engine



ACT 1 — SCALE

The six-stage cycle



No man steps in the same river twice
— it is not the same river, and he is not
the same man.

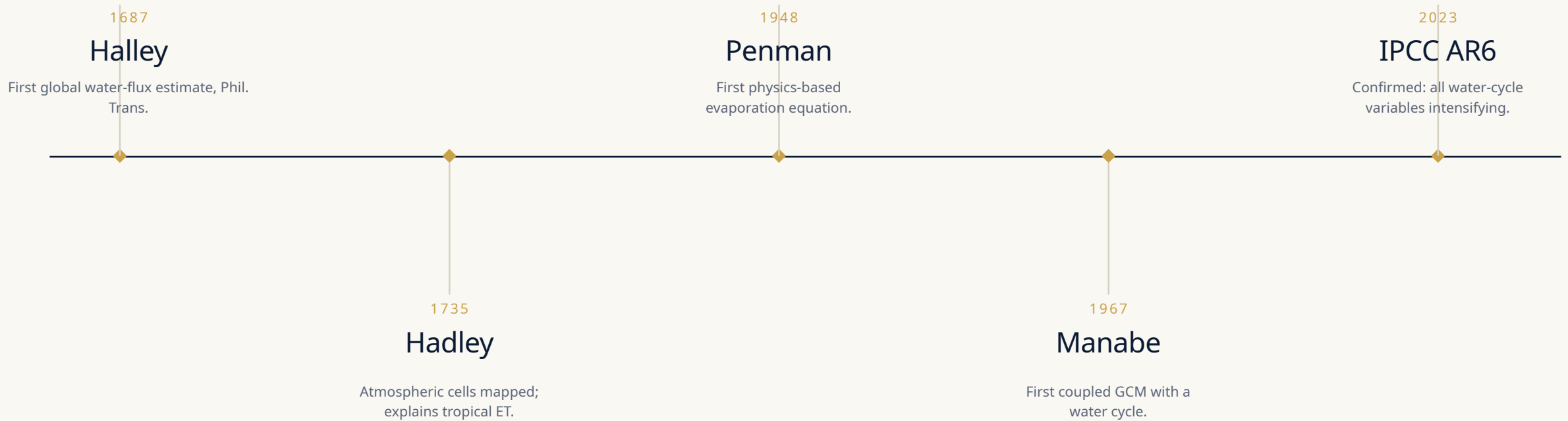
HERACLITUS, C. 500 BCE (DIELS-KRANZ 22 B 41)

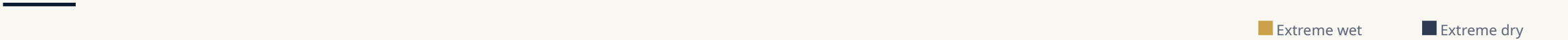
Changing faster than we measured for.

Act 2 — Stress

ACT 2 — STRESS

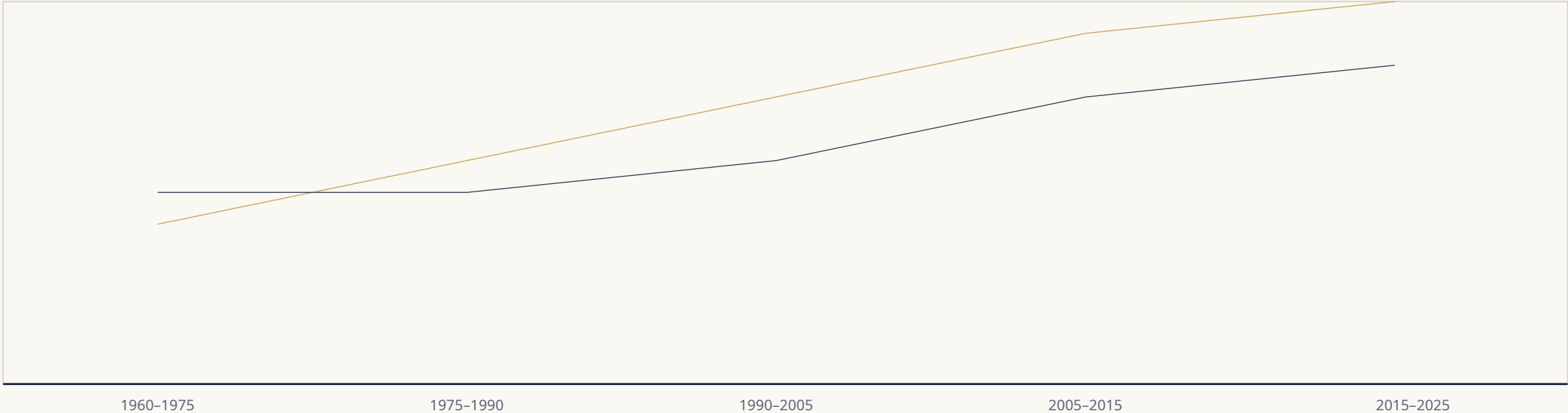
Three centuries of measurement





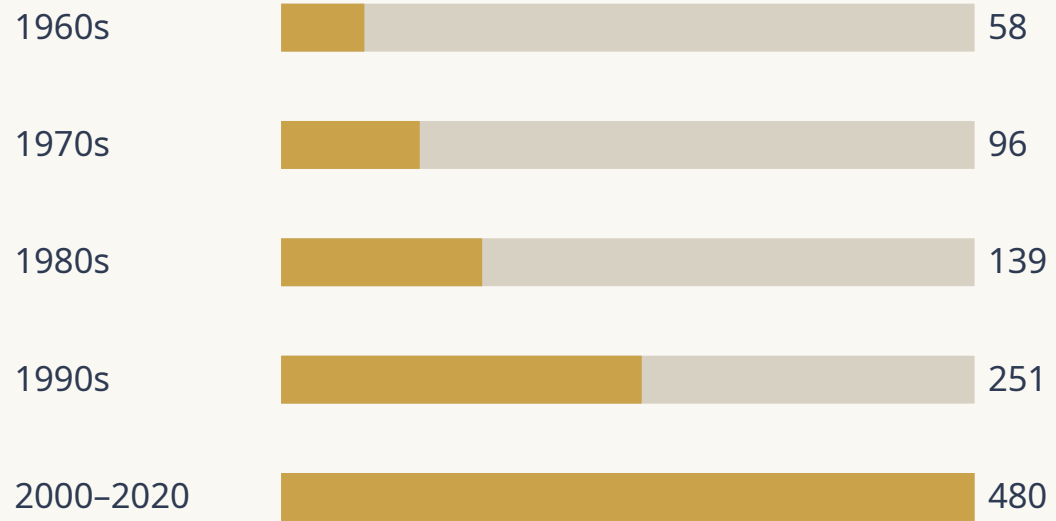
ACT 2 — STRESS

Precipitation extremes, 1960-2025



ACT 2 — STRESS

Floods, concentrated



Major flood events per decade — NOAA + EM-DAT



ACT 2 — STRESS

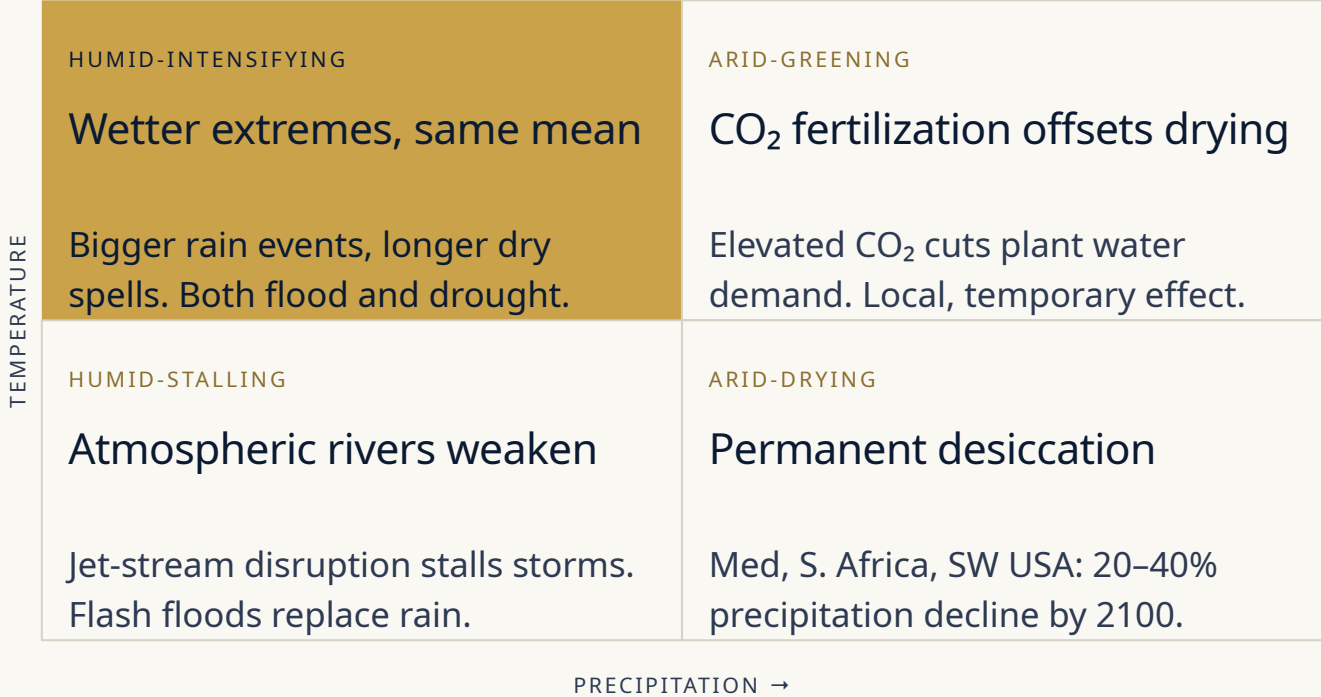
Water stress by region, 2025

ROW	COUNTRY	WSI	REGION	POP. AT RISK	PRIMARY DRIVER
01	Bahrain	5.0	Middle East	1.7M	No rivers; desalination
02	Cyprus	4.9	Med. Europe	1.2M	Summer drought
03	Kuwait	4.9	Middle East	4.3M	Arid; oil-for-water
04	Lebanon	4.8	Levant	5.5M	Governance collapse
05	Oman	4.8	Middle East	4.5M	Groundwater depletion
06	Qatar	4.8	Middle East	2.7M	Full desalination
07	UAE	4.7	Middle East	9.2M	Virtual water imports



ACT 2 — STRESS

Four climate regimes

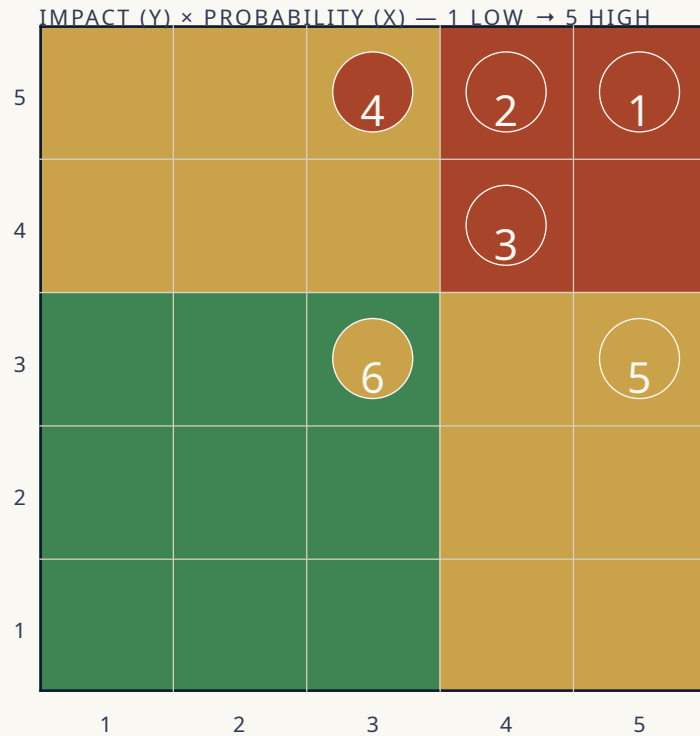


Reading the matrix

X axis: long-term precipitation trend (wetter → drier). Y axis: mean temperature rise. Most regions occupy Humid-Intensifying or Arid-Drying — both high-cost quadrants.

ACT 2 — STRESS

Water-cycle risks



RISKS

- 1 Groundwater depletion — non-renewable aquifer drawdown
- 2 Flood infrastructure failure — 100-yr events now 20-yr
- 3 Agricultural drought — staple crop failure, multi-season
- 4 Snowpack loss — seasonal water storage collapse
- 5 Urban heat-island amplifies local water stress
- 6 Saltwater intrusion into coastal freshwater lenses

ACT 2 — STRESS

Clausius– Clapeyron

Every 1°C of warming lets air hold 7% more water vapour. Wet regions get wetter, dry regions drier. At +2°C warming, rain extremes are 14% more intense.





ACT 2 — STRESS

What we measure

WORKSTREAM	INDICATOR	2000 BASELINE	2024 STATUS	TREND
Sea surface temp.	Global SST anomaly	+0.4°C vs 1981–2010	+0.77°C vs 1981–2010	Accelerating
Atm. moisture	Precipitable water	27.1 mm global mean	28.6 mm (+5.5%)	Upward
Snowpack	N. Hemisphere SWE	Declining since 1980	–15% vs 1981–2010	Declining
Glacier mass	WGMS global balance	–500 mm w.e./yr	–1050 mm w.e./yr	Accelerating loss

ACT 2 — STRESS

What measurement bought us

+4 days

LEAD TIME GAINED FOR MAJOR FLOOD WARNINGS SINCE 1990

67%

IMPROVEMENT IN 5-DAY PRECIPITATION FORECAST SKILL (ECMWF)



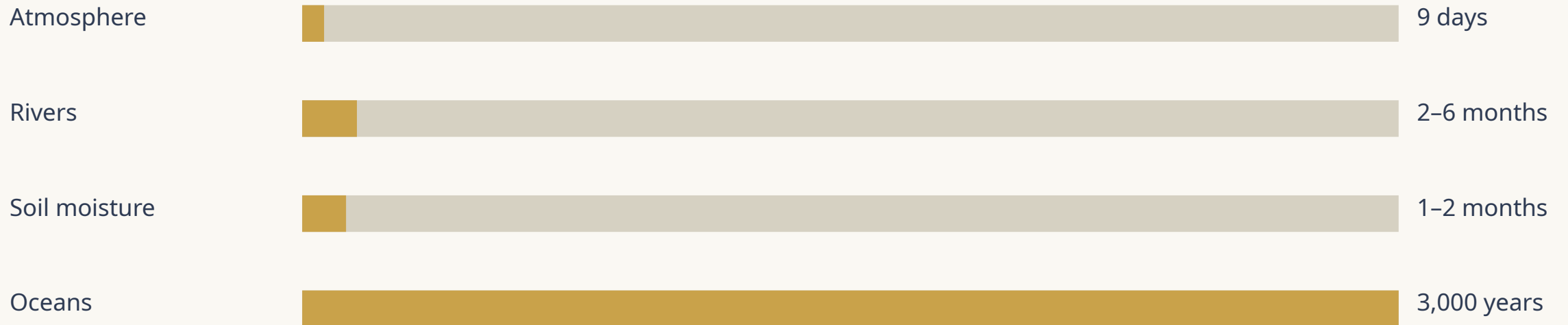


Six pathways.

Act 3 — Pathways

ACT 3 — PATHWAYS

Residence times

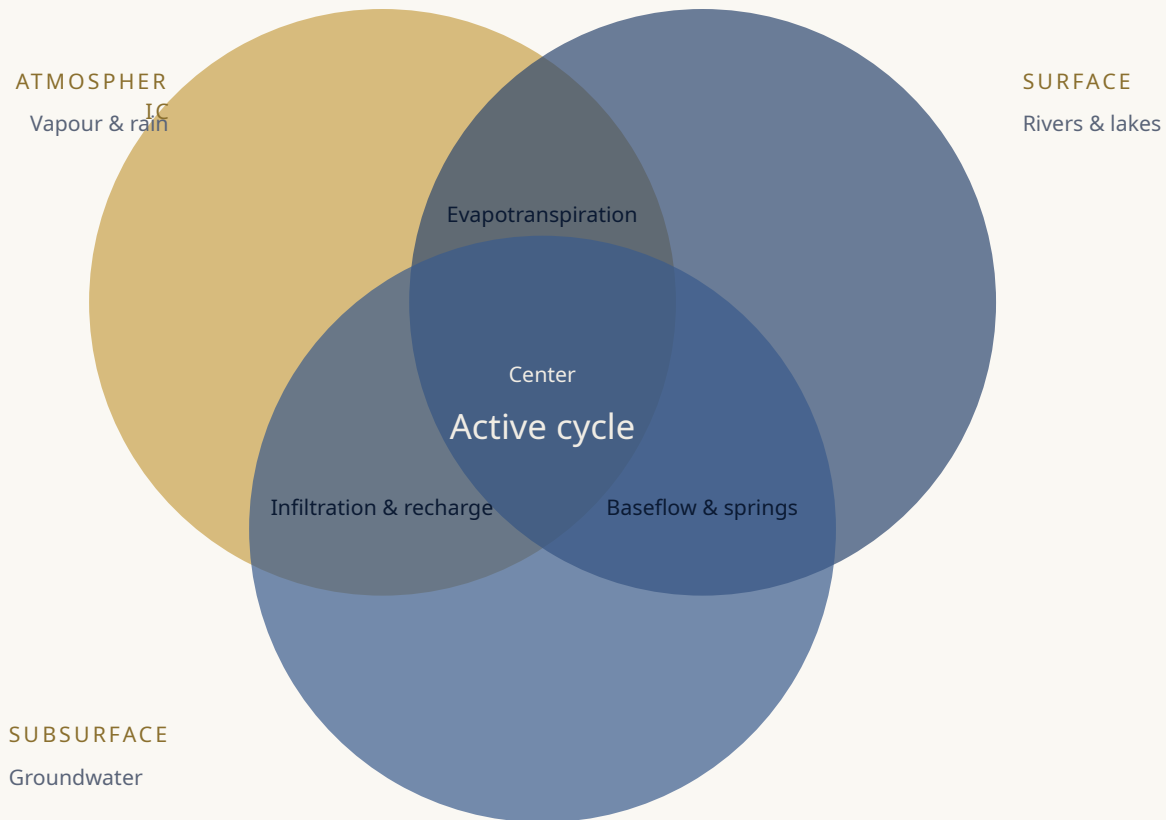


So what: Ocean water cycles slower than human civilisation; atmosphere recycles in 9 days.

USGS Water Science Glossary; Shiklomanov 1993

ACT 3 — PATHWAYS

Three phases overlap



A

Evapotranspiration

Plants return ~60% of terrestrial precipitation to atmosphere. Biosphere bridges surface and atmospheric phases.

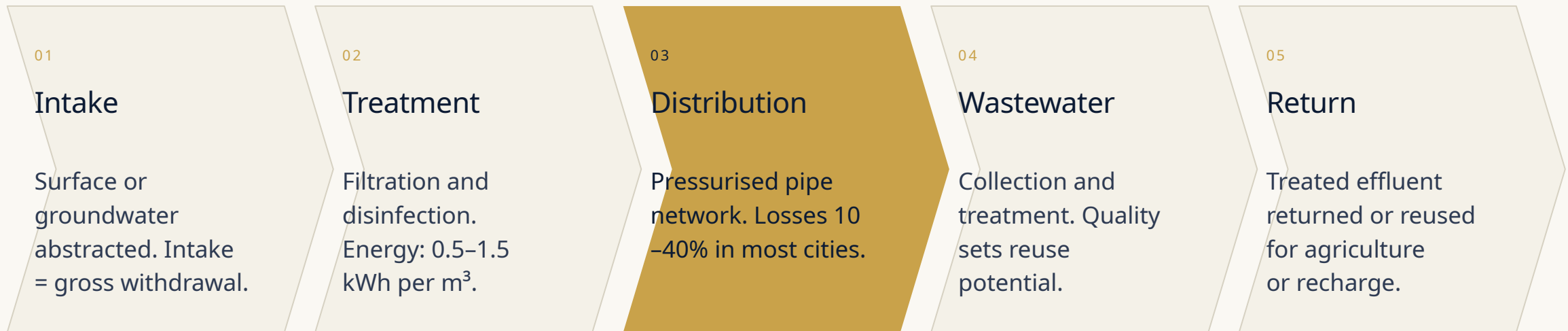
B

Groundwater recharge

Only ~3% of precipitation reaches aquifers. Deep systems recharge over millennia; pumping exceeds replenishment.

ACT 3 — PATHWAYS

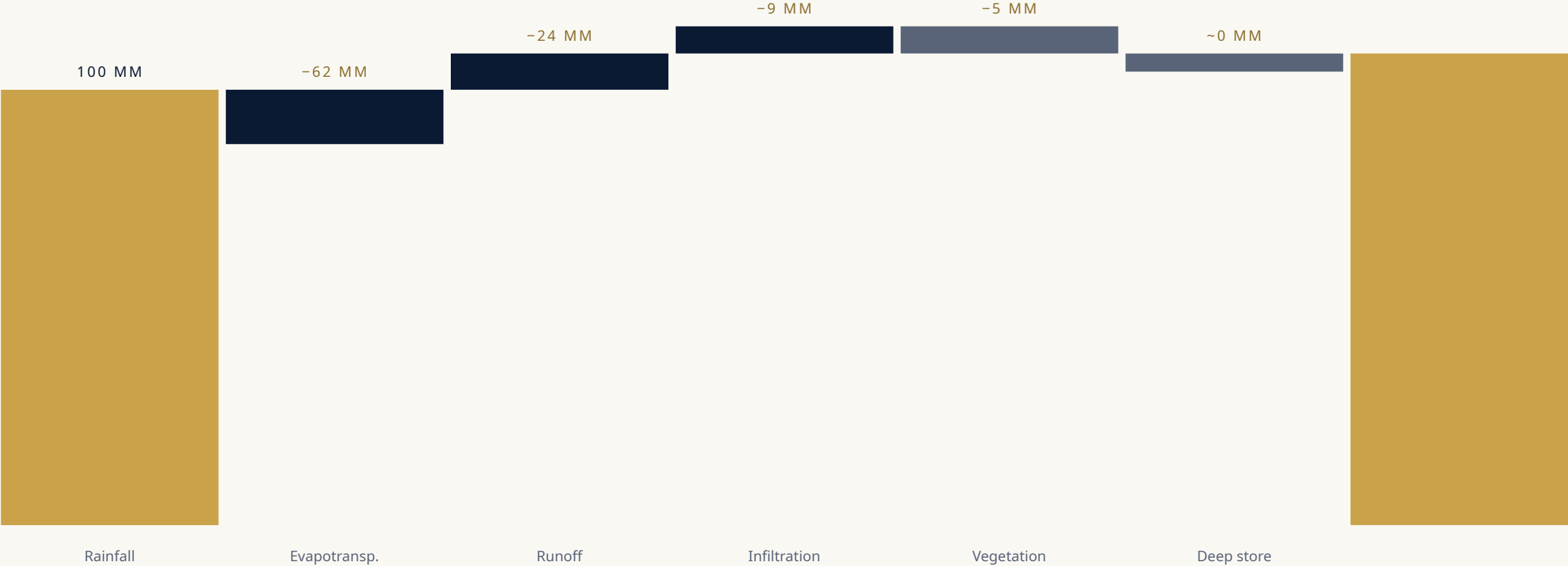
Urban water loop





ACT 3 — PATHWAYS

Fate of a rainfall





ACT 3 — PATHWAYS

From rain to glass



Rainfall
Total precipitation on catchment

100
mm^{100%}

Runoff
Reaches rivers and reservoirs

24 mm^{24%}

Captured
Stored in managed reservoirs

14 mm^{14%}

Treated
Enters drinking-water works

8 mm^{8%}

−76 mm
ET + UNCAPTURED RUNOFF

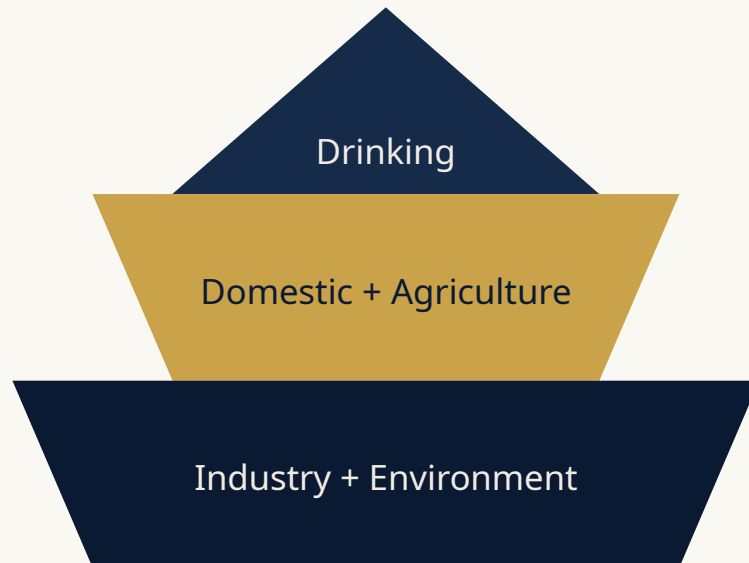
ET (62 mm) + runoff (14 mm) = 76% lost before treatment begins.

−6 mm
DISTRIBUTION LOSSES

Leaks and metering errors remove 6 mm. Global NRW average: 30–35%.

ACT 3 — PATHWAYS

Water-use hierarchy



ESSENTIAL HUMAN NEEDS

2–5 L/person/day minimum; 50 L/day for basic hygiene.
Protected under UN Resolution 64/292.

DOMESTIC & FOOD SECURITY

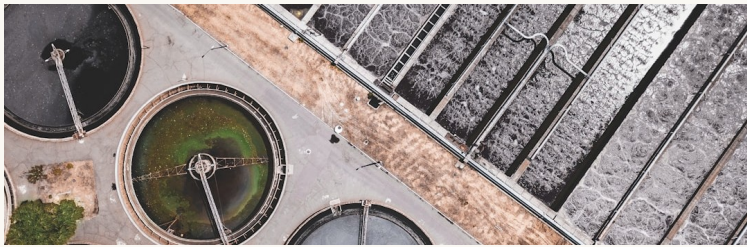
70% of global freshwater goes to irrigation; domestic adds
~15–20%. The food-water-energy nexus.

ECONOMIC PRODUCTION

Industry uses ~10% of withdrawals. Minimum environmental
flows protect ecosystem services.

ACT 3 — PATHWAYS

Six places doing it right



Singapore — closed loop
40% of supply from reclaimed wastewater



Israel — drip efficiency
90%+ drip irrigation; highest globally



Denmark — bog restoration
Rewetted peatlands store water + carbon



Netherlands — room for rivers
Floodplains absorb 1-in-1000-yr events



Kenya — watershed payment
Farmers paid to restore upstream forests



SF — purple pipe
Recycled water saves 10M gallons/day

Technology + policy + nature — none alone is sufficient.

ACT 4 — RESPONSE

PYRAMID PRINCIPLE CLOSE

Three levers that work

MOVE 01

Conserve demand at point of use

- Price water at full social cost
- Mandate drip irrigation above 100 ha
- Retrofit networks — cut NRW to <20%
- Label appliance water efficiency

30–40% demand reduction

Municipal utilities · Agriculture

MOVE 02

Reuse treated wastewater at scale

- Expand direct potable reuse globally
- Mandate industrial cooling-water loops
- Incentivise greywater recycling on-site
- Build purple-pipe in new developments

40% new supply in arid cities

Water utilities · Urban planners

MOVE 03

Restore catchments that regulate flow

- Rewet drained peatlands — 10M ha
- Reforest headwaters to restore baseflow
- Pay for ecosystem services at scale
- Reconnect floodplains; managed retreat

20–50% baseflow gain

Environmental agencies · Land planners

THREE THINGS TO REMEMBER

Same water · It's accelerating · We're a stage

01

Same water

Every molecule you drink has cycled through glaciers, deep aquifers, and ancient seas. The cycle is closed. No new water — only redistributed water.

USGS · Shiklomanov 1993

02

It's accelerating

Warming guarantees 7% more moisture per degree. Extremes intensify at both ends. Every key indicator has accelerated since 2000.

IPCC AR6 WG1 Ch.8

03

We're a stage

Humanity abstracts under 1% of annual flux. But we concentrate it in already-scarce places. Geography is the problem, not total quantity.

FAO AQUASTAT · WRI Aqueduct 4.0

ACT 4 — RESPONSE

Four water futures

Drier

Emissions push the Mediterranean, SW USA, and S. Africa toward aridification. Groundwater buffers exhausted by mid-century.



Wetter

Intensified monsoons bring catastrophic flooding to South and SE Asia — water abundance that cannot be stored or managed.



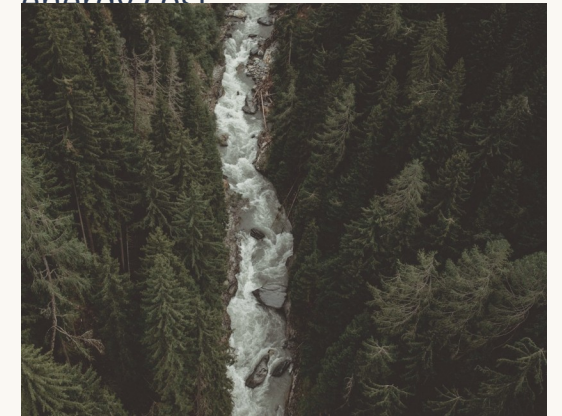
Managed

Desalination, advanced recycling, and real-time basin management keep cities supplied at high economic and energy cost.



Wild

Rewilded catchments and restored wetlands recover ecological function and regulate hydrological extremes at lower energy cost.





Same water.

IT'S ALL THE SAME WATER. · WATER-CYCLE SHOWCASE · MAY 2026